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Semiconductor device

Abstract

A semiconductor device includes a substrate, a light-receiving element, a switching element, a light-emitting element on the switching element, first and second conductive members. The substrate includes first to third metal pads on a front-side thereof. The light-receiving element includes first and second bonding pads on a front-surface thereof. A back-surface of the light-receiving element is connected to the first and second metal pads. The first and second bonding pads each overlap one of the first and second metal pads. The switching element includes front-side and backside electrodes and a control pad. The backside electrode is connected to the third metal pad. The first conductive member is connected to the front-side electrode of the switching element and the first bonding pad of the light-receiving element. The second conductive member is connected to the control pad of the switching element and the second bonding pad of the light-receiving element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-088345, filed on May 31, 2022; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments relate to a semiconductor device.

BACKGROUND

(3) Semiconductor devices are generally required to be downsized. A semiconductor device may have a structure, for example, in which multiple semiconductor elements are mounted on a substrate and subsequently resin-sealed on the substrate. Such a device preferably has a height reduced by using a thin substrate. However, discrepancies such as warp and like may occur when heating the thin substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic view showing a semiconductor device according to an embodiment;
- (2) FIGS. 2A and 2B are other schematic views showing the semiconductor device according to the embodiment;
- (3) FIGS. 3A to 3C are schematic views showing the substrate of the semiconductor device according to the embodiment;
- (4) FIG. 4 is a circuit diagram showing the semiconductor device according to the embodiment;
- (5) FIGS. 5A and 5B are schematic plan views showing substrates of the semiconductor device according to the embodiment;
- (6) FIGS. 6A and 6B are schematic plan views showing the substrate of the semiconductor device according to modifications of the embodiment;
- (7) FIGS. 7A and 7B are schematic plan views showing the substrate of the semiconductor device according to other modifications of the embodiment; and
- (8) FIG. 8 is a graph showing characteristics of the semiconductor device according to the embodiment.

DETAILED DESCRIPTION

(9) According to one embodiment, a semiconductor device includes a substrate, a light-receiving element, a switching element, a first conductive member, a second conductive member, and a light-emitting element. The substrate includes a resin base material having a first surface and first to third metal pads on the first surface. The first to third metal pads are arranged along the first surface. The third metal pad is apart from the first and second metal pads. The first surface includes a portion provided between the first and second metal pads. The light-receiving element is provided at the first surface side of the resin base material. The light-receiving element includes a front surface, a back surface, a first bonding pad and a second bonding pad. The back surface is provided at a side opposite to the front surface and connected via a first connection member to the first metal pad, the second metal pad, and the portion of the first surface of the resin base material. The first and second bonding pads are provided on the front surface of the light-receiving element. The first bonding pad overlaps one of the first and second metal pads in a first direction perpendicular to the first surface of the resin base material. The second bonding pad overlaps one of the first and second metal pads in the first direction. The switching element is provided at the first surface side of the resin base material, and includes a front-side electrode, a backside electrode and a control pad. The

front-side electrode is provided at a front surface side of the switching element. The backside electrode is provided at a back surface side opposite to the front surface side of the switching element and is connected to the third metal pad via a second connection member. The front-side electrode and the control pad are arranged at the front surface side of the switching element. The first conductive member is connected to the front-side electrode of the switching element and the first bonding pad of the light-receiving element. The first conductive member electrically connects the front-side electrode and the first bonding pad. The second conductive member is connected to the control pad of the switching element and the second bonding pad of the light-receiving element. The second conductive member electrically connects the control pad and the second bonding pad. The light-emitting element is provided on the front surface of the light-receiving element with a third connection member interposed. The third connection member transmits light radiated from the light-emitting element toward the light-receiving element.

(10) Embodiments will now be described with reference to the drawings. The same portions inside the drawings are marked with the same numerals; a detailed description is omitted as appropriate; and the different portions are described. The drawings are schematic or conceptual; and the relationships between the thicknesses and widths of portions, the proportions of sizes between portions, etc., are not necessarily the same as the actual values thereof. The dimensions and/or the proportions may be illustrated differently between the drawings, even in the case where the same portion is illustrated.

(11) There are cases where the dispositions of the components are described using the directions of XYZ axes shown in the drawings. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other. Hereinbelow, the directions of the X-axis, the Y-axis, and the Z-axis are described as an X-direction, a Y-direction, and a Z-direction. Also, there are cases where the Z-direction is described as upward and the direction opposite to the Z-direction is described as downward.

(12) FIG. 1 is a schematic view showing a semiconductor device 1 according to an embodiment. The semiconductor device 1 is, for example, a photo-relay. The semiconductor device 1 includes a substrate 10, a light-receiving element 20, a switching element 30, and a light-emitting element 40.

(13) The substrate 10 includes a resin base material 11, multiple metal pads 13, 15, and 17, an input-side terminal 50, and an output-side terminal 60. The resin base material 11 is, for example, sheet-like polyimide.

(14) The resin base material 11 includes a first surface 11F, and a second surface 11B at a side opposite to the first surface 11F. The multiple metal pads 13, 15, and 17 are provided on the first surface 11F. The input-side terminal 50 and the output-side terminal 60 are provided on the second surface 11B of the resin base material 11. The resin base material 11 is, for example, polyimide resin or epoxy resin.

(15) The multiple metal pads 13, 15, and 17 are apart from each other on the first surface 11F. Also, the input-side terminal 50 and the output-side terminal 60 are apart from each other on the second surface 11B. The multiple metal pads 13, 15, and 17, the input-side terminal 50, and the output-side terminal 60 are, for example, copper films provided with gold plating on the front surface.

(16) The light-receiving element 20, the switching element 30, and the light-emitting element 40 are provided at the first surface 11F side of the resin base material 11.

(17) The light-receiving element 20 includes a back surface 20BS connected to the metal pad 13 via a first connection member CM1, and a front surface 20FS at the side opposite to the back surface 20BS. The light-receiving element 20 further includes a first bonding pad 21 and a second bonding pad 23 provided on the front surface 20FS. The first connection member CM1 is, for example, a bonding material that includes a resin. The first connection member CM1 may be conductive or insulative.

(18) The switching element 30 includes a backside electrode 31, a front-side electrode 33 and a control pad 35. The backside electrode 31 is connected to the metal pad 15 via a second connection member CM2. The front-side electrode 33 is provided on the front surface at the side opposite to

the back surface at which the backside electrode **31** is provided. The control pad **35** is provided on the same front surface as the front-side electrode **33**. The second connection member **CM2** is, for example, a conductive bonding material such as a silver paste, etc.

(19) The front-side electrode **33** is electrically connected to the first bonding pad **21** of the light-receiving element **20** via a first conductive member **MW1**. The control pad **35** is electrically connected to the second bonding pad **23** of the light-receiving element **20** via a second conductive member **MW2**. The first conductive member **MW1** and the second conductive member **MW2** are, for example, metal wires.

(20) The light-emitting element **40** is connected via a third connection member **CM3** to the light-receiving element **20** at the front surface **20FS** side thereof. The third connection member **CM3** is, for example, a transparent resin and transmits light radiated from the light-emitting element **40** toward the light-receiving element **20**. The light-emitting element **40** is electrically connected to the metal pad **17** by a third conductive member **MW3**. The third conductive member **MW3** is, for example, a metal wire.

(21) The light-emitting element **40** is sealed on the front surface **20FS** of the light-receiving element **20** by a resin member **70**. The resin member **70** is, for example, silicone. Furthermore, the light-receiving element **20**, the switching element **30**, and the resin member **70** are sealed at the first surface **11F** side of the resin base material **11** by a resin member **80**. The resin member **80** is, for example, an epoxy resin or a silicone resin.

(22) FIGS. **2A** and **2B** are other schematic views showing the semiconductor device **1** according to the embodiment. FIG. **2A** is a perspective view showing the semiconductor device **1**. FIG. **2B** is a plan view showing the layout at the first surface **11F** side of the resin base material **11**. The resin members **70** and **80** are not illustrated in FIG. **2B**.

(23) As shown in FIG. **2A**, the semiconductor device **1** includes, for example, a first switching element **30a** and a second switching element **30b**. The first switching element **30a** and the second switching element **30b** each are electrically connected to the light-receiving element **20**.

(24) The resin member **80** seals the light-receiving element **20**, the switching element **30**, and the light-emitting element **40**. The conductive members **MW** and the metal pads **13**, **15**, and **17** provided on the first surface **11F** of the resin base material **11** also are sealed inside the resin member **80**. In other words, the components that are provided on the substrate **10** are housed inside a resin package that includes the resin base material **11** and the resin member **80**. In the following description, there may be a case in which the first switching element **30a** and the second switching element **30b** are referred to as the switching element **30** without distinction. The other components also described in the same manner.

(25) As shown in FIG. **2B**, the multiple metal pads **13**, **15**, and **17** include a first metal pad **13a**, a second metal pad **13b**, a third metal pad **15a**, a fourth metal pad **15b**, a fifth metal pad **17a**, and a sixth metal pad **17b**.

(26) The first metal pad **13a**, the third metal pad **15a**, and the fifth metal pad **17a** are arranged in a direction, e.g., an X-direction, along the first surface **11F**. The first metal pad **13a** is provided between the third metal pad **15a** and the fifth metal pad **17a**.

(27) The second metal pad **13b**, the fourth metal pad **15b**, and the sixth metal pad **17b** also are arranged, for example, in the X-direction. The second metal pad **13b** is provided between the fourth metal pad **15b** and the sixth metal pad **17b**.

(28) The first metal pad **13a** and the second metal pad **13b** are separated from each other and are arranged in a direction, e.g., a Y-direction, crossing the X-direction. The third metal pad **15a** and the fourth metal pad **15b** also are apart from each other and are arranged in, for example, the Y-direction. The fifth metal pad **17a** and the sixth metal pad **17b** also are apart from each other and are arranged in, for example, the Y-direction.

(29) The light-receiving element **20** is provided over the first metal pad **13a** and the second metal pad **13b**. The first surface **11F** includes a portion located between the first metal pad **13a** and the

second metal pad **13b**. The light-receiving element **20** is connected to the first metal pad **13a**, the second metal pad **13b**, and the portion of the first surface **11F** via the first connection member **CM1**. The light-receiving element **20** includes first bonding pads **21a** and **21b** and second bonding pads **23a** and **23b**. The first bonding pad **21a** and the second bonding pad **23a** are provided on the front surface of the light-receiving element **20** such that the first bonding pad **21a** and the second bonding pad **23a** overlap the first metal pad **13a** in a direction, e.g., a Z-direction, perpendicular to the first surface **11F** of the resin base material **11**. The first bonding pad **21b** and the second bonding pad **23b** also are provided on the front surface of the light-receiving element **20** such that the first bonding pad **21b** and the second bonding pad **23b** overlap the second metal pad **13b** in the Z-direction.

(30) The first switching element **30a** is provided on the third metal pad **15a** with the second connection member **CM2** interposed. The first switching element **30a** includes the backside electrode **31** (see FIG. 1), a front-side electrode **33a**, and a control pad **35a**. The backside electrode **31** is connected to the third metal pad **15a** via the second connection member **CM2**. The second connection member **CM2** is conductive; and the backside electrode **31** is electrically connected to the third metal pad **15a**.

(31) The front-side electrode **33a** is electrically connected to the first bonding pad **21a** of the light-receiving element **20** via the first conductive member **MW1**. The first conductive member **MW1** is, for example, a metal wire and is connected to the front-side electrode **33a** and the first bonding pad **21a**. The control pad **35a** is electrically connected to the second bonding pad **23a** of the light-receiving element **20** via the second conductive member **MW2**. The second conductive member **MW2** is, for example, a metal wire and is connected to the control pad **35a** and the second bonding pad **23a**.

(32) The second switching element **30b** is provided on the fourth metal pad **15b** with another second connection member **CM2** interposed. The second switching element **30b** includes a control pad **35b**, a front-side electrode **33b**, and another backside electrode **31**. The backside electrode **31** is electrically connected to the fourth metal pad **15b** via the second connection member **CM2**. The front-side electrode **33b** is electrically connected to the first bonding pad **21b** of the light-receiving element **20** via a fourth conductive member **MW4**. The control pad **35b** is electrically connected to the second bonding pad **23b** of the light-receiving element **20** via a fifth conductive member **MW5**. The fourth conductive member **MW4** and the fifth conductive member **MW5** are, for example, metal wires.

(33) The front-side electrode **33a** of the first switching element **30a** is electrically connected to the front-side electrode **33b** of the second switching element **30b** via a sixth conductive member **MW6**. The front-side electrode **33a** may be electrically connected to the front-side electrode **33b** via multiple sixth conductive members **MW6**. The sixth conductive member **MW6** is, for example, a metal wire.

(34) The light-emitting element **40** is provided on the front surface **20FS** of the light-receiving element **20** with the third connection member **CM3** interposed (see FIG. 1). The light-emitting element **40** is provided between the first bonding pads **21a** and **21b** and between the second bonding pads **23a** and **23b**.

(35) The light-emitting element **40** includes a back surface connected to the light-receiving element **20** via the third connection member **CM3**, and a front surface at the side opposite to the back surface. The light-emitting element **40** includes a first electrode **41** and a second electrode **43**. The first electrode **41** and the second electrode **43** are arranged on the front surface at the side opposite to the back surface. The first electrode **41** is electrically connected to the fifth metal pad **17a** via the third conductive member **MW3**. The second electrode **43** is electrically connected to the sixth metal pad **17b** via a seventh conductive member **MW7**. The seventh conductive member **MW7** is, for example, a metal wire.

(36) FIGS. 3A to 3C are schematic views showing the substrate **10** of the semiconductor device **1**

according to the embodiment. FIG. 3A is a plan view showing the second surface **11B** of the resin base material **11**. FIG. 3B is a cross-sectional view along line A-A shown in FIG. 3A. FIG. 3C is a cross-sectional view along line B-B shown in FIG. 3A.

(37) As shown in FIG. 3A, the semiconductor device **1** includes the multiple input-side terminals **50** and the multiple output-side terminals **60**. The multiple input-side terminals **50** and the multiple output-side terminals **60** are apart from each other. The input-side terminals **50** and the output-side terminals **60** are, for example, copper films provided with gold plating.

(38) The multiple input-side terminals **50** and the multiple output-side terminals **60** include, for example, a first input-side terminal **50a**, a second input-side terminal **50b**, a first output-side terminal **60a**, and a second output-side terminal **60b**. The first input-side terminal **50a** and the second input-side terminal **50b** are arranged, for example, in the Y-direction on the second surface **11B** of the resin base material **11**. The first output-side terminal **60a** and the second output-side terminal **60b** are arranged, for example, in the Y-direction on the second surface **11B** of the resin base material **11**. The first input-side terminal **50a** and the first output-side terminal **60a** are arranged, for example, in the X-direction. The second input-side terminal **50b** and the second output-side terminal **60b** are arranged, for example, in the X-direction.

(39) As shown in FIG. 3B, the resin base material **11** extends between the first input-side terminal **50a** and the fifth metal pad **17a**. The resin base material **11** extends between the first output-side terminal **60a** and the third metal pad **15a**. The first input-side terminal **50a** is electrically connected to the fifth metal pad **17a** by a via contact BC1 provided in the resin base material **11**. The first output-side terminal **60a** is electrically connected to the third metal pad **15a** by a via contact BC2 provided in the resin base material **11**. Moreover, the multiple via contacts BC2 may electrically connects the first output-side terminal **60a** and the third metal pad **15a**.

(40) The resin base material **11** includes a portion of the second surface **11B** located between the first input-side terminal **50a** and the first output-side terminal **60a**. The first metal pad **13a** is provided at the side opposite to the portion of the second surface **11B** of the resin base material **11**. The first input-side terminal **50a** includes, for example, a portion overlapping the first metal pad **13a** in the Z-direction via the resin base material **11**. The first output-side terminal **60a** includes, for example, a portion overlapping the first metal pad **13a** in the Z-direction via the resin base material **11**. In other words, the first metal pad **13a** is provided via the resin base material **11** on an edge of the first input-side terminal **50a** and an edge of the first output-side terminal **60a**.

(41) As shown in FIG. 3C, the resin base material **11** extends between the second input-side terminal **50b** and the sixth metal pad **17b**. Also, the resin base material **11** extends between the second output-side terminal **60b** and the fourth metal pad **15b**. The second input-side terminal **50b** is electrically connected to the sixth metal pad **17b** by a via contact BC3 provided in the resin base material **11**. The second output-side terminal **60b** is electrically connected to the fourth metal pad **15b** by a via contact BC4 provided in the resin base material **11**. Moreover, the multiple via contacts BC4 may electrically connects the second output-side terminal **60b** and the fourth metal pad **15b**.

(42) The resin base material **11** includes a portion of the second surface **11B** located between the second input-side terminal **50b** and the second output-side terminal **60b**. The second metal pad **13b** is provided at the side opposite to the portion of the second surface **11B** of the resin base material **11**. The second input-side terminal **50b** includes, for example, a portion overlapping the second metal pad **13b** in the Z-direction via the resin base material **11**. The second output-side terminal **60b** includes, for example, a portion overlapping second metal pad **13b** in the Z-direction via the resin base material **11**. In other words, the second metal pad **13b** is provided via the resin base material **11** on an edge of the second input-side terminal **50b** and an edge of the second output-side terminal **60b**.

(43) FIG. 4 is a circuit diagram showing the semiconductor device **1** according to the embodiment. The light-receiving element **20** is, for example, a photodiode. The first switching element **30a** and

the second switching element **30b** are, for example, MOS transistors. The light-emitting element **40** is, for example, a light-emitting diode.

(44) The light-receiving element **20** includes multiple photodiodes **25** and a control circuit **27**. The multiple photodiodes **25** are connected in series. The photodiodes **25** are configured to detect the light of the light-emitting element **40**. The control circuit **27** is, for example, a waveform shaping circuit. The control circuit **27** may include a discharging circuit, a protection circuit and like.

(45) The output of the photodiodes **25** is output to the first bonding pads **21a** and **21b** and the second bonding pads **23a** and **23b** via the control circuit **27**. The first bonding pads **21a** and **21b** are electrically connected to, for example, the cathode side of the photodiodes **25**. The second bonding pads **23a** and **23b** are electrically connected to, for example, the anode side of the photodiodes **25**.

(46) The first bonding pad **21a** of the light-receiving element **20** is electrically connected to a source S (the front-side electrode **33a**) of the switching element **30a** via the first conductive member MW1. The first bonding pad **21b** of the light-receiving element **20** is electrically connected to the source S (the front-side electrode **33b**) of the switching element **30b** via the fourth conductive member MW4. The source S of the switching element **30a** is connected to the source S of the switching element **30b** via the sixth conductive member MW6.

(47) The second bonding pad **23a** of the light-receiving element **20** is electrically connected to a gate G (the control pad **35a**) of the first switching element **30a** via the second conductive member MW2. The second bonding pad **23b** of the light-receiving element **20** is electrically connected to the gate G (the control pad **35b**) of the second switching element **30b** via the fifth conductive member MW5.

(48) The light-emitting element **40** is electrically connected to the first input-side terminal **50a** via the third conductive member MW3. The first input-side terminal **50a** is electrically connected to the fifth metal pad **17a** by the via contact BC1 (see FIG. 3B); and the third conductive member MW3 is bonded to the fifth metal pad **17a** and the first electrode **41** of the light-emitting element **40** (see FIG. 2B). The first electrode **41** is electrically connected to, for example, the cathode of the light-emitting diode.

(49) Also, the light-emitting element **40** is electrically connected to the second input-side terminal **50b** via the seventh conductive member MW7. The second input-side terminal **50b** is electrically connected to the sixth metal pad **17b** by the via contact BC3 (see FIG. 3B); and the seventh conductive member MW7 is bonded to the sixth metal pad **17b** and the second electrode **43** of the light-emitting element **40**. The second electrode **43** is electrically connected to, for example, the anode of the light-emitting diode.

(50) The first output-side terminal **60a** is electrically connected to a drain D (the backside electrode **31**) of the first switching element **30a**. The first output-side terminal **60a** is electrically connected to the third metal pad **15a** by the via contact BC2 (see FIG. 3A); and the backside electrode **31** of the first switching element **30a** is electrically connected to the third metal pad **15a** via the second connection member CM2 (see FIG. 2B).

(51) The second output-side terminal **60b** is electrically connected to the drain D (the other backside electrode **31**) of the second switching element **30b**. The second output-side terminal is electrically connected to the fourth metal pad **15b** by the via contact BC4 (see FIG. 3B); and the backside electrode **31** of the second switching element **30b** is electrically connected to the fourth metal pad **15b** via the other second connection member CM2 (see FIG. 2B).

(52) For example, the on-off control signal is input to the first and second input-side terminals **50a** and **50b** to control the electrical conduction between the first output-side terminal **60a** and the second output-side terminal **60b**. The light-emitting element **40** radiates an optical signal corresponding to the signal input to the first and second input-side terminals **50a** and **50b**; and the light-receiving element **20** detects the optical signal radiated from the light-emitting element **40**. The light-receiving element **20** outputs a control signal corresponding to the optical signal to the gates G of the first and second switching elements **30a** and **30b**. The electrical conduction between

the first output-side terminal **60a** and the second output-side terminal **60b** is controlled thereby. (53) For example, when transmitting a high-frequency signal from the first output-side terminal **60a** to the second output-side terminal **60b**, it is preferable to reduce the transmission loss by reduce the impedance between the first output-side terminal **60a** and the second output-side terminal **60b**. In the semiconductor device **1**, the impedances between the switching element **30a** and the first output-side terminal **60a** and between the switching element **30b** and the second output-side terminal **60b** are reduced by reducing the Z-direction thickness of the resin base material **11**. The pass characteristics of the high-frequency signal between the first output-side terminal **60a** and the second output-side terminal **60b** can be improved thereby.

(54) The Z-direction thickness of the resin base material **11** is, for example, 50 micrometers. The Z-direction thicknesses of the metal pads **13**, **15**, and **17** are, for example, 30 micrometers; and the Z-direction thicknesses of the input-side terminal **50** and the output-side terminal **60** are, for example, 30 micrometers. Thus, the Z-direction thickness of the resin base material **11** is preferably less than the combined thickness of the copper films provided respectively on the first surface **11F** and on the second surface **11B**. However, when such a thin resin base material **11** is used in the substrate **10**, for example, warp or deformation is generated in the resin base material **11** by thermal stress due to the difference between the linear thermal expansion coefficients of the resin base material **11** and the copper films. Therefore, in the manufacturing processes of the semiconductor device **1**, a mechanical failure may occur due to the degraded flatness of the resin base material **11**.

(55) As shown in FIGS. **3B** and **3C**, the input-side terminals **50** face the metal pads **17** via the resin base material **11**. The output-side terminals **60** face the metal pads **15** via the resin base material **11**. The thermal stress in the resin base material **11** due to the different linear thermal expansion coefficients is cancelled and reduced between the terminal and the metal pad facing each other via the resin base material **11**. However, the thermal stress is not canceled at the second surface **11B** side of the resin base material **11** that faces the metal pads **13** because a copper film is not provided. Therefore, warp and/or deformation caused by the metal pads **13** easily occurs in the substrate **10**.

(56) In contrast, in the semiconductor device **1**, the thermal stress is reduced by providing multiple metal pads **13** on the resin base material **11** so that the area occupied by the multiple metal pads **13** is reduced. Moreover, the first surface **11F** of the resin base material **11** includes a portion located between the multiple metal pads **13**; and the connection strength between the substrate **10** and the light-receiving element **20** can be increased by connecting the light-receiving element **20** and the resin base material **11** at the portion of the first surface **11F** via the first connection member **CM1**. In other words, the connection strength between the substrate **10** and the first connection member **CM1** can be increased by the bonding between resins in addition to the bonding between resin and metal.

(57) Moreover, the first and second bonding pads **21** and **23** of the light-receiving element **20** are provided at positions where the first and second bonding pads **21** and **23** overlap the metal pads **13** in the Z-direction. Thereby, the absorption of the ultrasonic waves by the resin base material **11** can be suppressed while performing ultrasonic bonding of the conductive members **MW** on the bonding pads **21** and **23**, and the bonding strength of the conductive members **MW** can be increased.

(58) As shown in FIGS. **3A** to **3C**, the metal pad **13** includes a portion overlapping the input-side terminal **50** or the output-side terminal **60**. The absorption or scattering of the ultrasonic waves by the resin base material **11** can be further reduced thereby. Moreover, the substrate **10** may be placed on a support surface while mounting the light-receiving element **20** on the metal pad **13**, but the second surface **11B** of the resin base material **11** located between the input terminal **50** and the output terminal **60** does not come into contact with the support surface. Therefore, the resin base material **11** may be deformed by stress due to the pressure applied to the metal pad **13**. Such deformation of the resin base material **11** can be suppressed by the metal pad **13** partially overlapping the input terminal **50** or the output terminal **60**. Further, the metal pad **13** may overlap

the input-side terminal **50** and the output-side terminal **60**.

(59) Moreover, the first electrode **41** and the second electrode **43** of the light emitting element **40** preferably overlap the metal pad **13** in a plan view parallel to the first surface **11F** of the resin base material **11**. Thereby, it is possible to stably bond the conductive members **MW3** and **MW7** to the first electrode **41** and the second electrode **43** and increase the respective bonding strengths thereof.

(60) FIGS. **5A** and **5B** are schematic plan views showing substrates **10** of the semiconductor device **1** according to the embodiment. FIG. **5A** is a plan view showing the first surface **11F** of the resin base material **11**. FIG. **5B** is a plan view showing the first surface **11F** of another substrate **10**. The broken lines in the drawings illustrate the positions of the light-receiving element **20** and the light-emitting element **40**.

(61) As shown in FIG. **5A**, the light-receiving element **20** is provided over the first metal pad **13a** and the second metal pad **13b**. The light-emitting element **40** also is provided over the first metal pad **13a** and the second metal pad **13b** on the light-receiving element **20**.

(62) The spacing between the first metal pad **13a** and the second metal pad **13b** is substantially equal to the spacing between the third metal pad **15a** and the fourth metal pad **15b**. The embodiment is not limited to the example; the spacing between the first metal pad **13a** and the second metal pad **13b**, for example, may be greater than the spacing between the third metal pad **15a** and the fourth metal pad **15b**. Also, the space between the first metal pad **13a** and the second metal pad **13b** may have uniform spacing as shown in FIG. **5A**. Alternatively, the spacing may be different at the center from the spacing at the ends in the X-direction.

(63) As shown in FIG. **5B**, the first metal pad **13a** may be linked to the second metal pad **13b** via a metal connection part **13c**. The first surface **11F** of the resin base material **11A** includes a portion surrounded with the first metal pad **13a**, the second metal pad **13b**, and two metal connection parts **13c**.

(64) The metal connection part **13c** links the outer edge of the first metal pad **13a** facing the third metal pad **15a** and the outer edge of the second metal pad **13b** facing the fourth metal pad **15b** on the first surface **11F** of the resin base material **11**.

(65) Another metal connection part **13c** links the outer edge of the first metal pad **13a** facing the fifth metal pad **17a** and the outer edge of the second metal pad **13b** facing the sixth metal pad **17b**.

(66) The first connection member **CM1** that connects the substrate **10** and the light-receiving element **20** is preferably a conductive bonding material such as a silver paste or the like that provides a large bonding force with respect to metal. However, the first connection member **CM1** may spread outward from the space between the metal pads **13** after being coated onto the first surface **11F** of the resin base material **11**. Thus, there may be a case where the spacing between the first connection member **CM1** and the metal pads **15** is narrow, and the breakdown voltage is reduced between the metal pads **13** and the metal pads **15**. In the example, the metal connection part **13c** prevents the first connection member **CM1** from spreading outward from the space between the metal pads **13**. Thereby, it is possible to prevent the reduction of the breakdown voltage between the metal pads **13** and the metal pads **15**.

(67) FIGS. **6A** and **6B** are schematic plan views showing the substrate **10** of the semiconductor device **1** according to modifications of the embodiment. FIGS. **6A** and **6B** are plan views showing the first surface **11F** of the resin base material **11**. The broken lines in the drawings illustrate the positions of the light-receiving element **20** and the light-emitting element **40**.

(68) In the example shown in FIG. **6A**, the substrate **10** further includes a seventh metal pad **13d** and an eighth metal pad **13e**. The seventh metal pad **13d** and the eighth metal pad **13e** are provided on the first surface **11F** of the resin base material **11**.

(69) The seventh metal pad **13d** is provided between the first metal pad **13a** and the fifth metal pad **17a**. The seventh metal pad **13d** is apart from the first and fifth metal pads **13a** and **17a**.

(70) The eighth metal pad **13e** is provided between the second metal pad **13b** and the sixth metal pad **17b**. The eighth metal pad **13e** is apart from the second and sixth metal pads **13b** and **17b**. The

seventh metal pad **13d** and the eighth metal pad **13e** are arranged in the Y-direction and apart from each other.

(71) The spacings between the metal pads may be the same or different. The spaces between the metal pads may have uniform widths or may have non-uniform widths.

(72) Thus, the thermal stress applied to the resin base material **11** can be reduced by providing the four metal pads **13** each having a small occupied area. Also, the light-receiving element **20** may have large connection strength to the substrate **10** provided by increasing the surface area of the first surface **11F** of the resin base material **11** between the metal pads **13** that are next to each other.

(73) In the Z-direction perpendicular to the first surface **11F** of the resin base material **11**, the first bonding pad **21a** of the light-receiving element **20** overlaps the first metal pad **13a**, and the second bonding pad **23a** overlaps the seventh metal pad **13d**. Also, the first bonding pad **21b** of the light-receiving element **20** overlaps the second metal pad **13b** in the Z-direction. The second bonding pad **23b** overlaps the eighth metal pad **13e** in the Z-direction. Moreover, the light-emitting element **10** is preferably mounted on the light-receiving element **40** such that four corners of the light-emitting element **10** overlap the metal pads **13a**, **13b**, **13d** and **13e**, respectively.

(74) In the example shown in FIG. **6B**, the metal connection parts **13c**, **13f**, **13g**, and **13h** are further included in the metal pad **13**. The metal connection parts **13c**, **13f**, **13g**, and **13h** are provided on the first surface **11F** of the resin base material **11** to surround the portion of the first surface **11F** located between the first metal pad **13a** and the second metal pad **13b**, between the first metal pad **13a** and the seventh metal pad **13d**, between the second metal pad **13b** and the eighth metal pad **13e**, and between the seventh metal pad **13d** and the eighth metal pad **13e**.

(75) The metal connection part **13c** links the outer edge of the first metal pad **13a** facing the third metal pad **15a** and the outer edge of the second metal pad **13b** facing the fourth metal pad **15b**.

(76) The metal connection part **13f** links the outer edge of the seventh metal pad **13d** facing the fifth metal pad **17a** and the outer edge of the eighth metal pad **13e** facing the sixth metal pad **17b**. The metal connection part **13g** links the outer edge along the X-direction of the first metal pad **13a** and the outer edge along the X-direction of the seventh metal pad **13d**. The metal connection part **13h** links the outer edge along the X-direction of the second metal pad **13b** and the outer edge along the X-direction of the eighth metal pad **13e**. Thereby, the metal connection parts **13c**, **13f**, **13g**, and **13h** prevent the first connection member **CM1** from spreading outward from the space between the multiple metal pads **13**.

(77) In FIG. **6B**, the light-receiving element **20** also is provided such that the first bonding pad **21a** overlaps the first metal pad **13a** and the second bonding pad **23a** overlaps the seventh metal pad **13d** in the Z-direction perpendicular to the first surface **11F** of the resin base material **11**. Moreover, the first and second bonding pads **21b** and **23b** of the light-receiving element **20** overlap the second metal pad **13b** and the eighth metal pad **13e**, respectively.

(78) The light-emitting element **40** is mounted on the light-receiving element **20** such that the four corners of the light-emitting element **40** overlap the metal pads **13a**, **13b**, **13d** and **13e**, respectively. Moreover, the first and second electrodes **41** and **43** of the light-emitting element **40** preferably overlap the metal pad **13** in a plan view parallel to the first surface **11F** of the resin base material **11**. Thereby, the conductive members **MW3** and **MW7** are stably bonded to the light-emitting element **40**, and the bonding strengths thereof are increased.

(79) FIGS. **7A** and **7B** are schematic plan views showing the substrate **10** of the semiconductor device **1** according to other modifications of the embodiment. FIGS. **7A** and **7B** are plan views showing the first surface **11F** of the resin base material **11**. The broken lines in the drawings illustrate the positions of the light-receiving element **20** and the light-emitting element **40**.

(80) As shown in FIG. **7A**, the second metal pad **13b** is apart from the first metal pad **13a** and surrounds the first metal pad **13a**. The first metal pad **13a** is, for example, circular.

(81) The light-emitting element **40** is provided at a position where the light-emitting element **40** overlaps the first metal pad **13a** in the Z-direction. The first bonding pads **21a** and **21b** and the

second bonding pads **23a** and **23b** of the light-receiving element **20** are provided at positions where the bonding pads overlap the second metal pad **13b** in the Z-direction.

(82) In the example shown in FIG. 7B, the second metal pad **13b** also is apart from the first metal pad **13a** and surrounds the first metal pad **13a**. The first metal pad **13a** is, for example, triangular. The light-emitting element **40** is provided at a position where the light-emitting element **40** overlaps the first metal pad **13a** in the Z-direction. The first bonding pads **21a** and **21b** and the second bonding pads **23a** and **23b** of the light-receiving element **20** are provided at positions where the bonding pads overlap the second metal pad **13b** in the Z-direction.

(83) In these examples, the light-receiving element **20** also is provided over the first metal pad **13a** and the second metal pad **13b**. By increasing the surface area of the first surface **11F** of the resin base material **11** located between the first metal pad **13a** and the second metal pad **13b**, the occupied areas of the first and second metal pads **13a** and **13b** can be reduced, and it is possible to reduce the thermal stress in the resin base material **11** due to the different thermal expansion coefficients of the materials of the first and second metal pads **13a** and **13b** and the resin base material **11**. The warp or deformation of the resin base material **11** can be suppressed thereby. Also, the light-receiving element **20** may have large connection strength to the substrate **10**. The first metal pad **13a** surrounded with the second metal pad **13b** may have a shape not limited to the example above, and may be, for example, polygonal.

(84) Moreover, in FIGS. 7A and 7B, the first electrode **41** and the second electrode **43** also preferably overlap the metal pad **13** in a plan view parallel to the first surface **11F** of the resin base material **11**. Thereby, the conductive members MW3 and MW7 are stably bonded to the light-emitting element **40**, and the bonding strengths thereof are increased.

(85) FIG. 8 is a graph showing characteristics of the semiconductor device **1** according to the embodiment. The horizontal axis is the frequency of the signal propagating between the first output-side terminal **60a** and the second output-side terminal **60b**. The vertical axis is the insertion loss between the first output-side terminal **60a** and the second output-side terminal **60b**.

(86) “EB” shown in FIG. 8 illustrates the characteristic of the semiconductor device **1**. “CE” illustrates the characteristic of a semiconductor device according to a comparative example. In the semiconductor device according to the comparative example, the first metal pad **13a** and the second metal pad **13b** (see FIG. 5A) are provided as a continuous body.

(87) As shown in FIG. 8, the frequency band of the semiconductor device **1** is wider than the frequency band of the semiconductor device according to the comparative example. In other words, the frequency characteristics can be improved by dividing the metal pad **13**, on which the light-receiving element **20** is mounted, into the first and second metal pads **13a** and **13b**.

(88) In recent years, semiconductor devices such as memory elements and the like operate with higher speeds, and thus, the high speed operation is an urgent issue for the photo-relays used in testers thereof. As shown in FIG. 8, in the semiconductor device **1** according to the embodiment, high-frequency characteristics up to the 30 GHz band are achieved by thinning the substrate **10** and downsizing the entire semiconductor device **1**. Furthermore, it can be seen that the signal propagates from the first output-side terminal **60a** to the second output-side terminal **60b** or in the opposite direction under the -3 dB bandwidth widened from 25 GHz to 30 GHz by mutually-separating the multiple metal pads on which the light-receiving element **20** is mounted. It is considered that such an effect can be achieved by the parasitic capacitance parallel to the signal path between the first output-side terminal **60a** and the second output-side terminal **60b** that is drastically reduced by subdividing the mount pad.

(89) Embodiments include the following aspects:

Note 1

(90) A semiconductor device, comprising: a substrate including a resin base material having a first surface and first to third metal pads on the first surface, the first to third metal pads being arranged along the first surface, the third metal pad being apart from the first and second metal pads, the first

surface including a portion provided between the first and second metal pads; a light-receiving element provided at the first surface side of the resin base material, the light-receiving element including a front surface, a back surface, a first bonding pad and a second bonding pad, the back surface being provided at a side opposite to the front surface, the back surface being connected via a first connection member to the first metal pad, the second metal pad, and the portion of the first surface of the resin base material, the first and second bonding pads being provided on the front surface of the light-receiving element, the first bonding pad overlapping one of the first and second metal pads in a first direction perpendicular to the first surface of the resin base material, the second bonding pad overlapping one of the first and second metal pads in the first direction; a switching element provided at the first surface side of the resin base material, the switching element including a front-side electrode, a backside electrode and a control pad, the front-side electrode being provided at a front surface side of the switching element, the backside electrode being provided at a back surface side opposite to the front surface side of the switching element and being connected to the third metal pad via a second connection member, the front-side electrode and the control pad being arranged at the front surface side of the switching element; a first conductive member connected to the front-side electrode of the switching element and the first bonding pad of the light-receiving element, the first conductive member electrically connecting the front-side electrode and the first bonding pad; a second conductive member connected to the control pad of the switching element and the second bonding pad of the light-receiving element, the second conductive member electrically connecting the control pad and the second bonding pad; and a light-emitting element provided on the front surface of the light-receiving element with a third connection member interposed, the third connection member transmitting light radiated from the light-emitting element toward the light-receiving element.

Note 2

(91) The device according to note 1, wherein the second metal pad is apart from the first metal pad.

Note 3

(92) The device according to note 1, wherein the substrate further includes a metal connection part provided on the first surface of the resin base material, the metal connection part linking the first metal pad and the second metal pad, and the first metal pad, the second metal pad, and the metal connection part surround the portion of the first surface of the resin base material.

Note 4

(93) The device according to any one of notes 1 to 3, further comprising another switching element provided at the first surface side of the resin base material and electrically connected to the light-receiving element, wherein the substrate further includes a fourth metal pad provided on the first surface of the resin base material, the fourth metal pad being apart from the first to third metal pads, the first metal pad and the second metal pad are arranged in a second direction along the first surface of the resin base material, the third and fourth metal pads also being arranged in the second direction, and said another switching element is provided on the fourth metal pad via another second connection member.

Note 5

(94) The device according to note 1, wherein the substrate further includes fifth and sixth metal pads provided on the first surface of the resin base material, the fifth and sixth metal pads being arranged in a second direction along the first surface of the resin base material, the first and second metal pads are arranged in the second direction, the first metal pad being positioned between the third metal pad and the fifth metal pad, the second metal pad being positioned between the third metal pad and the sixth metal pad, the resin base material includes the portion of the first surface extending between the first metal pad and the second metal pad, between the first metal pad and the fifth metal pad, between the second metal pad and the sixth metal pad, and between the fifth metal pad and the sixth metal pad, the light-receiving element has the back surface connected via the first connection member to the first metal pad, the second metal pad, the fifth metal pad, the sixth metal

pad, and the portion of the first surface of the resin base material, the first bonding pad of the light-receiving element overlaps one of the first metal pad, the second metal pad, the fifth metal pad and the sixth metal pad in the first direction, and the second bonding pad of the light-receiving element overlaps one of the first metal pad, the second metal pad, the fifth metal pad and the sixth metal pad in the first direction.

Note 6

(95) The device according to note 5, wherein the first metal pad, the second metal pad, the fifth metal pad, and the sixth metal pad are apart from each other.

Note 7

(96) The device according to note 5, wherein the substrate further includes a first metal connection part provided between the first metal pad and the second metal pad, a second metal connection part provided between the second metal pad and the sixth metal pad, a third metal connection part provided between the fifth metal pad and the sixth metal pad, and a fourth metal connection part provided between the fifth metal pad and the first metal pad, the first to fourth metal connection parts being arranged to surround the portion of the first surface.

Note 8

(97) The device according to note 1, wherein the second metal pad is apart from the first metal pad and surrounds the first metal pad.

Note 9

(98) The device according to note 8, wherein the light-emitting element overlaps the first metal pad in the first direction perpendicular to the first surface of the resin base material.

Note 10

(99) The device according to note 9, wherein the first bonding pad and the second bonding pad of the light-receiving element overlap the second metal pad in the first direction perpendicular to the first surface of the substrate.

Note 11

(100) The device according to any one of notes 1 to 10, wherein the substrate further includes an input-side terminal and an output-side terminal provided on the resin base material, wherein the resin base material has a second surface opposite to the first surface, the input-side terminal and the output-side terminal being provided on the second surface of the resin base material, wherein the input-side terminal is electrically connected to the light-emitting element, and the output-side terminal is electrically connected to the switching element via the third metal pad and the second connection member.

(101) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and overview of the invention.

Claims

1. A semiconductor device, comprising: a substrate including a resin base material having a first surface and first to third metal pads on the first surface, the first to third metal pads being arranged along the first surface, the third metal pad being apart from the first and second metal pads, the first surface including a portion provided between the first and second metal pads; a light-receiving element provided at the first surface side of the resin base material, the light-receiving element including a front surface, a back surface, a first bonding pad and a second bonding pad, the back surface being provided at a side opposite to the front surface, the back surface being connected via

a first connection member to the first metal pad, the second metal pad, and the portion of the first surface of the resin base material, the first and second bonding pads being provided on the front surface of the light-receiving element, the first bonding pad overlapping one of the first and second metal pads in a first direction perpendicular to the first surface of the resin base material, the second bonding pad overlapping one of the first and second metal pads in the first direction; a switching element provided at the first surface side of the resin base material, the switching element including a front-side electrode, a backside electrode and a control pad, the front-side electrode being provided at a front surface side of the switching element, the backside electrode being provided at a back surface side opposite to the front surface side of the switching element and being connected to the third metal pad via a second connection member, the front-side electrode and the control pad being arranged at the front surface side of the switching element; a first conductive member connected to the front-side electrode of the switching element and the first bonding pad of the light-receiving element, the first conductive member electrically connecting the front-side electrode and the first bonding pad; a second conductive member connected to the control pad of the switching element and the second bonding pad of the light-receiving element, the second conductive member electrically connecting the control pad and the second bonding pad; and a light-emitting element provided on the front surface of the light-receiving element with a third connection member interposed, the third connection member transmitting light radiated from the light-emitting element toward the light-receiving element.

2. The device according to claim 1, wherein the second metal pad is apart from the first metal pad.

3. The device according to claim 1, wherein the substrate further includes a metal connection part provided on the first surface of the resin base material, the metal connection part linking the first metal pad and the second metal pad, and the first metal pad, the second metal pad, and the metal connection part surround the portion of the first surface of the resin base material.

4. The device according to claim 1, wherein the light-emitting element partially overlaps the first metal pad and the second metal pad in the first direction.

5. The device according to claim 1, further comprising: another switching element provided at the first surface side of the resin base material and electrically connected to the light-receiving element, the substrate further including a fourth metal pad provided on the first surface of the resin base material, the fourth metal pad being apart from the first to third metal pads, the first metal pad and the second metal pad being arranged in a second direction along the first surface of the resin base material, the third and fourth metal pads also being arranged in the second direction, said another switching element being provided on the fourth metal pad via another second connection member.

6. The device according to claim 5, wherein a first space provided between the first metal pad and the second metal pad is aligned in a third direction with a second space provided between the third metal pad and the fourth metal pad, the third direction crossing the second direction.

7. The device according to claim 1, wherein the substrate further includes fifth and sixth metal pads provided on the first surface of the resin base material, the fifth and sixth metal pads being arranged in a second direction along the first surface of the resin base material, the first and second metal pads are arranged in the second direction, the first metal pad being positioned between the third metal pad and the fifth metal pad, the second metal pad being positioned between the third metal pad and the sixth metal pad, the first surface of the resin base material includes the portion extending between the first metal pad and the second metal pad, between the first metal pad and the fifth metal pad, between the second metal pad and the sixth metal pad, and between the fifth metal pad and the sixth metal pad, the light-receiving element has the back surface connected via the first connection member to the first metal pad, the second metal pad, the fifth metal pad, the sixth metal pad, and the portion of the first surface of the resin base material, the first bonding pad of the light-receiving element overlaps one of the first metal pad, the second metal pad, the fifth metal pad and the sixth metal pad in the first direction, and the second bonding pad of the light-receiving element overlaps one of the first metal pad, the second metal pad, the fifth metal pad and the sixth metal

pad in the first direction.

8. The device according to claim 7, wherein a first space provided between the first metal pad and the second metal pad is aligned in a third direction with a third space provided between the fifth metal pad and the sixth metal pad, the third direction crossing the second direction, and a fourth space provided between the first metal pad and the fifth metal pad is arranged in the second direction with a fifth space provided between the second metal pad and the sixth metal pad.

9. The device according to claim 7, wherein the first metal pad, the second metal pad, the fifth metal pad, and the sixth metal pad are apart from each other.

10. The device according to claim 7, wherein the substrate further includes a first metal connection part provided between the first metal pad and the second metal pad, a second metal connection part provided between the second metal pad and the sixth metal pad, a third metal connection part provided between the fifth metal pad and the sixth metal pad, and a fourth metal connection part provided between the fifth metal pad and the first metal pad, the first to fourth metal connection parts being arranged to surround the portion of the first surface.

11. The device according to claim 1, wherein the second metal pad is apart from the first metal pad and surrounds the first metal pad.

12. The device according to claim 11, wherein the light-emitting element overlaps the first metal pad in the first direction perpendicular to the first surface of the resin base material.

13. The device according to claim 12, wherein the first bonding pad and the second bonding pad of the light-receiving element overlap the second metal pad in the first direction perpendicular to the first surface of the substrate.

14. The device according to claim 1, wherein the substrate further includes an input-side terminal and an output-side terminal provided on the resin base material, the resin base material having a second surface opposite to the first surface, the input-side terminal and the output-side terminal being provided on the second surface of the resin base material, the input-side terminal being electrically connected to the light-emitting element, the output-side terminal being electrically connected to the switching element via the third metal pad and the second connection member.

15. The device according to claim 14, wherein the input-side terminal includes first and second input terminal parts arranged along the second surface of the resin base material, the first input terminal part being apart from the second input terminal part, the substrate further includes a seventh metal pad and an eighth metal pad provided on the first surface of the resin base material, the seventh metal pad being electrically connected to the first input terminal part, the eighth metal pad being electrically connected to the second input terminal part, and the light-emitting element is electrically connected to the first input terminal via the seventh metal pad and electrically connected to the second input terminal via the eighth metal pad.

16. The device according to claim 15, wherein the first metal pad is provided between the third metal pad and the seventh metal pad, and the second metal pad is provided between the third metal pad and the eighth metal pad.
